

Ko'p o'zgaruvchili funktsiyaning xususiy hosilalari

Reja

- 1°. Funktsiyaning xususiy hosilalari tushunchasi.
- 2°. Ko'p o'zgaruvchili funktsiyaning differensiallanuvchiligi. Zaruriy shart.
- 3°. Funktsiya differensiallanuvchiligining yetarli sharti

1°. Funktsiyaning xususiy hosilalari tushunchasi

Faraz qilaylik, $f(x) = f(x_1, x_2, \dots, x_m)$ funksiya $E \subset R^m$ to'plamda berigan bo'lib,

$$x^0 = (x_1^0, x_2^0, \dots, x_m^0) \in E, \quad (x_1^0 + \Delta x_1, x_2^0, \dots, x_m^0) \in E \quad (\Delta x_1 \geq 0)$$

bo'lsin. Bu funktsiyaning x^0 nuqtadagi x_1 o'zgaruvchi bo'yicha xususiy ortirmasi

$$\Delta_{x_1} f(x^0) = f(x_1^0 + \Delta x_1, x_2^0, \dots, x_m^0) - f(x_1^0, x_2^0, \dots, x_m^0)$$

Δx_1 ga bog'liq bo'ladi.

1-ta'rif. Ushbu

$$\lim_{\Delta x_1 \rightarrow 0} \frac{\Delta_{x_1} f(x^0)}{\Delta x_1}$$

limit mavjud bo'lsa, bu limit $f(x) = f(x_1, x_2, \dots, x_m)$ funktsiyaning $x^0 = (x_1^0, x_2^0, \dots, x_m^0)$ nuqtadagi x_1 o'zgaruvchisi bo'yicha xususiy hosilasi deyiladi.

Uni

$$\frac{\partial f(x^0)}{\partial x_1} \text{ yoki } f'_{x_1}(x^0)$$

kabi belgilanadi:

$$\begin{aligned} \frac{\partial f(x^0)}{\partial x_1} &= f'_{x_1} = \lim_{\Delta x_1 \rightarrow 0} \frac{\Delta_{x_1} f(x^0)}{\Delta x_1} = \\ &= \lim_{\Delta x_1 \rightarrow 0} \frac{f(x_1^0 + \Delta x_1, x_2^0, \dots, x_m^0) - f(x_1^0, x_2^0, \dots, x_m^0)}{\Delta x_1}. \end{aligned}$$

Berilgan funktsiyaning bu xususiy hosilasini quyidagicha

$$f'_{x_1}(x^0) = \lim_{x_1 \rightarrow x_1^0} \frac{f(x_1, x_2^0, \dots, x_m^0) - f(x_1^0, x_2^0, \dots, x_m^0)}{x_1 - x_1^0}$$

ta'riflana ham bo'ladi.

Xuddi shunga o'xshash $f(x_1, x_2, \dots, x_m)$ funksiyaning boshqa $x_2, x_3, \dots, x_m$ o'zgaruvchilari bo'yicha xususiy hosilalari ta'riflanadi:

$$\begin{aligned} \frac{\partial f(x^0)}{\partial x_2} &= \lim_{\Delta x_2 \rightarrow 0} \frac{\Delta_{x_2} f(x^0)}{\Delta x_2} = \\ &= \lim_{\Delta x_2 \rightarrow 0} \frac{f(x_1^0, x_2^0 + \Delta x_2, x_3^0, \dots, x_m^0) - f(x_1^0, x_2^0, \dots, x_m^0)}{\Delta x_2}, \dots \\ \frac{\partial f(x^0)}{\partial x_m} &= \lim_{\Delta x_m \rightarrow 0} \frac{\Delta_{x_m} f(x^0)}{\Delta x_m} = \\ &= \lim_{\Delta x_m \rightarrow 0} \frac{f(x_1^0, x_2^0, \dots, x_{m-1}^0, x_m^0 + \Delta x_m) - f(x_1^0, x_2^0, \dots, x_m^0)}{\Delta x_m}. \quad (\Delta x_k > 0, k = 2, 3, \dots, m) \end{aligned}$$

Yuqorida keltirilgan ta'riflardan ko'p o'zgaruvchili funksiyaning xususiy hosilalari bir o'zgaruvchili funksiyaning hosilasi kabi ekanligi ko'rinadi. Demak, ko'p o'zgaruvchili funksiyaning xususiy hosilalarini topishda ma'lum jadval va qoidalardan foydalanish mumkin. Jumladan, agar

funksiyalar $E \subset R^m$ to'plamda berilgan bo'lib, $x \in E$ nuqtada xususiy hosilalarga ega bo'lsa, u holda:

Misollar

Quyidagi funksiyalarning xususiy hosilalari topilsin.

1. a) $z = x^3 + y^2 - 2xy$, b) $z = 5x^2 + 8xy^2 + y^3$

2. a) $z = x^3 + y^2 - 3axy$, b) $z = x^2 - 2xy + y^2$

3. a) $z = \frac{xy}{x+y}$, b) $z = \frac{x-y}{x+y}$

4. a) $z = \frac{y}{x}$, b) $z = \sqrt{\frac{x}{y}}$